

atomOS

ASA operator engine



Blend operator & pinion integration

A-W- ϕ -N / hollow-center aperture

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This specification	ASA 3.1.0 — minor revision
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Integrated in this release

The circle-plus blend; coordinate A; time interval W; kinematic calculus B in the source's double-lowercase-phi notation; north polar pinion N; sweeping-envelope flags; and the hollow-center / active-fringe distinction.

Self-contained revision of the formal specification. The retained ASA/NA core, new definitions, compatibility conditions and source references are set out below. The five newly supplied source pages are reproduced in the appendix.

Release record

ASA 3.1.0 integrates the new blend-operator document into the existing ASA engine specification. It preserves the 3.0.0 aperture/NA/absorption core and adds a named pinion record plus an optional, explicitly supplied fringe mask. The new material is not treated as an unrelated replacement engine.

The source document proceeds from the A–W–B–N blend notation to a sweeping-cone interpretation, then to a boresight-aligned hollow center and active caustic fringe. That order is retained in this revision. [C, pp. 1–5]

Release item	Disposition
Unchanged core	ASA intersection, NA after ASA, POPCNT boundary count, whole-word Vantablack disposition.
New definitions	Typed A, W, B/ ϕ , N; one-bit blend interpretation; three separately named predicates.
New integration	Record-only default and optional supplied-mask profile, with an exact neutral-mask compatibility rule.
Unchanged storage	Logical word groups, tail-bit convention, canonical result order, existing linear/Morton interpretation.
Delivery	Formalized PDF, source mapping, algebra checks and original-source appendix.
Implementation status	Specification update only; no executable or CUDA revision is delivered or reported as tested here.

How to distinguish the material

C denotes the newly supplied five-page blend document. **A** denotes the earlier WHITE KING corpus; **B** denotes the fly-eye addendum. **B0** denotes the ASA 3.0.0 mathematical contract established in the preceding project conversation and restated here. It is a baseline definition, not a newly inspected binary artifact.

I01–I08 identify integration decisions made for this revision. They are not presented as numerical rules already supplied by C. A source quotation or source mapping remains distinct from a normative definition and from a deduction proved from that definition.

Minor-version compatibility

Existing 3.0 records keep their original meaning. By default, the new pinion information is a separate annotation. Enabling an additional fringe mask is explicit; selecting the neutral mask recovers the original core output exactly.

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Reading routes

For the integration itself, read Sections 3–8. For the source’s changing use of the one-bit state, read Section 5. For compatibility and review, read Sections 8–12. The appendix preserves C’s original wording and tables so that the formal interpretation can be checked against the supplied pages.

1 The retained ASA core

1.1 Typed word contract

For a fixed logical cell dictionary, let $x, M_{\text{ASA}}, M_{\text{NA}}, M_{\partial}, V \in \{0, 1\}^{32}$. They respectively denote the input occupancy, ASA aperture, post-ASA NA mask, boundary mask and valid-bit mask. The retained record is

$$a = x \wedge M_{\text{ASA}} \wedge V, \quad (1)$$

$$n = a \wedge M_{\text{NA}}, \quad (2)$$

$$h = \text{popcount}(n \wedge M_{\partial}), \quad (3)$$

$$y = \begin{cases} 0, & h > 0, \\ n, & h = 0, \end{cases} \quad F_{3.0}(x; M_{\text{ASA}}, M_{\text{NA}}, M_{\partial}, V) = (a, n, h, y). \quad (4)$$

This is the 3.0 contract used by the new integration. The NA stage remains after ASA, and any intersecting boundary bit zeros the entire selected logical word. [B0, staged word rule][A, pp. 75–81]

Whole-word disposition is not replaced by $n \wedge \neg M_{\partial}$. A one-bit region annotation is likewise not silently substituted for the boundary word. Those would be different operations with different output records.

1.2 Retained geometry and grouping

The baseline uses a declared log-polar dictionary, with $\rho = \ln(r/r_{\min})$ on a finite positive-radius annulus and a periodic angular index. A 32-bit word groups 32 logical angular predicates. A last partial word is masked by V ; padded storage does not create extra cells. The coordinate key, occupancy word and author identifier remain separate types. [B0, log-polar dictionary and padding]

The existing Morton profile interleaves radial *word-coordinate* bits in even positions and angular-word bits in odd positions within an 8-by-8 tile. Linear or Morton storage must not change which logical cells share a whole-word disposition. This revision leaves that grouping and its output order intact. [B0, word-level swizzle][A, pp. 85–87]

1.3 What stays outside the new blend

The double-UU author/state binding, symbolic genesis, facet-input organization and previously declared timing targets are inherited, not redefined by the letters A, W, B or N in C. The new W is a local time-interval label; it is not a reinterpretation of the author's date. The new N is a pinion datum; it is not the pre-existing NA mask. [I01]

2 The new document, in its own order

The five source pages are integrated through three successive concepts, not flattened into one unnamed Boolean gate.

C page	Source content	Integration in 3.1.0
1	Circle-plus operator, coordinate A, W as delta time, B described as double lowercase phi, and N as north polar pinion.	Preserve all names; assign distinct record fields and require a declared encoding before applying bitwise XOR.
2	$N \oplus (A + \phi(W)) \bmod 2$; zero as aligned and one as phase adjustment. Then the cone-sweep interpretation begins.	Retain the displayed expression; define its binary interpretation separately from continuous or geometric inputs.
3	A mapping table: N as datum/apex, A as boresight, (ϕ, W) as sweep parameters. Zero means inside the sweep envelope; one means limit/threshold event.	Preserve the table's relationships and create a distinct envelope predicate. The source's hardware actions remain described actions, not automatic record semantics.
4	Hollow-center / caustic receiver configuration; A becomes the centerline of a central void; inner and outer boundaries are distinguished.	Add named region roles, with geometry and boundary values left as supplied-profile inputs.
5	Zero for the hollow region and one for active fringe. The structural comparison contrasts boresight, receiver location and latch meaning.	Add a separate fringe predicate; retain the listed configuration alternatives and their unresolved choice.

2.1 The two source tables

C p.3 maps symbols to roles in the sweeping-parabola narrative. C p.5 changes that narrative to a hollow-center arrangement. Both tables are preserved in the appendix. Their entries are treated as *source configuration descriptions*; the PDF does not add a reflector design or infer a numerical reception pattern from those descriptions alone.

C closes with questions about pencil versus fan beam, waveform type, reflector versus ring-array configuration, and carrier range. No answers to those questions occur in the five-page file. They remain open profile fields rather than chosen hardware settings. [C, pp.3, 5]

3 A–W–B–N notation without name collisions

Source	Retained meaning	Canonical record name
A	Primary eigenvector axis; later boresight centerline.	A_{axis} / <code>axis_reference</code>
W	Infinitesimal or discrete time interval used by the kinematic calculus.	W / <code>time_interval</code>
B	Kinematic calculus described as double lowercase phi.	B_{ϕ} / <code>kinematic_descriptor</code>
$\phi(W)$	The source's notation for the accumulated shift or transformation over W .	<code>kinematic_result</code> , with its type declared.
N	North polar pinion; reference datum and later mechanical apex description.	N_{pinion} / <code>pinion_reference</code>
$\oplus$	Circle-plus blend; binary XOR/state-toggle interpretation.	<code>blend_xor</code> , operating on bits only.

[C, pp. 1–3][I01]

3.1 The double-lowercase-phi label

C uses the phrase “double lowercase phi” but displays a single ϕ in its explanatory formulas. This edition retains that phrase as the name of B_{ϕ} . It does not silently replace it with $\ddot{\phi}$, ϕ^2 , a second derivative, or an ordered pair. A meaning among those alternatives is not fixed by the supplied page. [C, p. 1]

The same paragraph calls B an angular velocity *or* a kinematic transformation matrix. Those are alternative types, not interchangeable arithmetic operands. Consequently, $\phi(W)$ remains a named result of the chosen B -profile; no integration equation or matrix is invented for it. [I02]

3.2 Axis, origin and code are distinct

Calling A an eigenvector axis identifies its source role. The underlying linear operator, eigenvalue, normalization and coordinate-frame convention are not specified. Likewise, a north direction, a mechanical origin and a pinion label need distinct representation when a later profile uses them numerically.

The previous ASA mask is written M_{ASA} , the NA mask M_{NA} , and the boundary mask M_{∂} . None is overwritten by the new source letters. This namespacing is required to retain the 3.0 word contract while incorporating C's notation. An encoded bit for an axis is a record value about that axis, not the entire axis itself.

4 The circle-plus blend: source expression and typing

4.1 Expression retained from C

The source writes

$$s_{\text{source}} = N \oplus (A + \phi(W)) \pmod{2}. \quad (5)$$

Its accompanying interpretation assigns zero to alignment and one to a phase-shift/toggle condition. [C, p. 2]

As written, the expression does not specify how a reference axis, a pinion datum and a continuous kinematic result become binary operands. The formalization therefore separates the *source expression* from a defined *binary encoding profile*. That distinction is part of the integration, not an implicit correction of C. [I02]

4.2 Defined binary special case

When a selected profile supplies $a_b, n_b, \beta_b \in \{0, 1\}$, define

$$s_{\text{blend}} = n_b \oplus ((a_b + \beta_b) \pmod{2}) = n_b \oplus a_b \oplus \beta_b. \quad (6)$$

Here β_b is the profile's encoded kinematic contribution, not W itself. The eight cases are

n_b	a_b	β_b	s_{blend}
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	1

The equality in Eq. (6) follows because addition modulo two and XOR have the same two-input truth table.

4.3 Meaning of equality after encoding

A zero result establishes equality of the encoded bit classes. It establishes equality of the original geometric states only if the encoding profile supplies that stronger implication. Parity alone cannot do so: the distinct integers zero and two have the same remainder modulo two.

For geometric inputs, a profile must provide its encoder and units before Eq. (6) is evaluated. When that information is absent, the blend value is *unspecified*; it is not silently replaced by zero. C supplies no such encoding rule or overflow convention. [C, pp. 1–2]

5 Three one-bit predicates, three meanings

The source uses the same words “State 0” and “State 1” for three different conditions. Version 3.1 preserves each polarity but stores the conditions separately. [I03]

Predicate	Source meaning of 0	Source meaning of 1	Page
s_{align}	Alignment / no phase adjustment.	Out-of-phase / phase-shift or toggle condition.	C 2
s_{limit}	Within the valid sweeping envelope.	Sweep limit or a detection threshold reached.	C 3
s_{fringe}	Inside the hollow / central void.	On the active caustic fringe.	C 4–5

5.1 The required independence

No equation in C establishes

$$s_{\text{align}} = s_{\text{limit}} = s_{\text{fringe}}. \quad (7)$$

The integration therefore does not make that identification. A change from zero to one in the alignment record is not automatically a fringe acceptance, and a fringe bit is not automatically a whole-word boundary count.

The binary value in Eq. (6) may populate s_{align} only under a profile that explicitly adopts C’s alignment interpretation. The other two predicates require their own supplied conditions. The source proposes hardware consequences for some states, but those consequences do not define equality between the predicates. [C, pp. 2–5]

5.2 Representing an unspecified value

Each predicate is paired with a definition status:

$$Q = (v, d), \quad d \in \{\text{defined}, \text{unspecified}\}, \quad v \in \{0, 1\} \text{ only when } d = \text{defined}. \quad (8)$$

This is a *typed record about a one-bit predicate*; the status is metadata, not a third value secretly placed in a one-bit word. It distinguishes a genuine source zero from a missing definition. [I03]

C p. 5 contrasts the hollow region with the active fringe, but does not assign a complete truth table for points beyond the outer boundary, points on an exact edge, or a missing observation. Those cases remain unspecified until a named profile defines them. This avoids treating “outside the hollow” as a synonym for “on the active fringe” without an explicit domain restriction.

6 Sweeping envelope and hollow-center aperture

6.1 The source's two-stage geometry narrative

C pp. 2–3 describes a parabolic detection arrangement with N as datum/apex, A as boresight and (ϕ, W) as the time-dependent sweep description. C pp. 4–5 then introduces a central receiver and distinguishes the hollow region from an active fringe around it. The second description refines the source scenario; it does not supply numerical dimensions for the first one.

C names Cassegrain, Gregorian and ring-focus alternatives, and later asks whether the configuration is a dual-reflector system or a phased ring array. No choice is finalized in the uploaded document. The specification retains the alternatives rather than silently choosing one. [C, pp. 4–5]

6.2 Configuration sets, not an inferred field

For documentary integration, let a supplied region description name H as the hollow region and R as the active fringe. On explicitly classified points, C 's polarity is represented as

$$s_{\text{fringe}}(z) = \begin{cases} 0, & z \in H, \\ 1, & z \in R. \end{cases} \quad (9)$$

A consistent binary profile requires $H \cap R = \emptyset$. If the profile does not classify z , the value is unspecified rather than inferred from this two-case definition. [I04]

C alternates between “two angles” and “inner/outer caustic radius.” This release keeps `inner_boundary` and `outer_boundary` as named, unit-tagged profile fields. It does not equate a radius with an angle, invent a focal length, or infer a ring-shaped response solely from receiver location. The source supplies a proposed hollow-center configuration but no field equation establishing those numerical relationships. [C, p. 4]

6.3 Connection to the ASA aperture

When a profile explicitly compiles its active-fringe classification into the existing logical cell dictionary, its support can be represented by M_{fringe} . That mask is an optional input to Section 7; it is not computed by an unstated radar or acoustic model.

The pinion and geometry record remains separate from the author record and from the facet source label. The fly-eye organization is retained as prior input context, not converted into a newly specified multi-sensor acquisition protocol. [B0, facet input model][B, pp. 7–8]

7 Integration into the ASA engine

7.1 Default profile: record-only

Let P_C be the pinion record containing the source references, A–W–B–N fields, definition statuses and the three named predicates. The default minor-version extension is

$$F_{3.1}^{\text{record}}(x; M, P_C) = \langle F_{3.0}(x; M), P_C \rangle, \quad (10)$$

where $M = (M_{\text{ASA}}, M_{\text{NA}}, M_{\partial}, V)$. Projection onto the core record gives the identical 3.0 result. Existing consumers can therefore read the original four fields without reinterpreting the new flags. [I05]

This is an additive integration: the new source model becomes part of the engine’s formal state record, while its unresolved parameters do not erase previously valid output.

7.2 Optional profile: explicitly supplied fringe mask

A caller may opt into an already-defined active-fringe mask over the same logical dictionary. The effective aperture is

$$M'_{\text{ASA}} = M_{\text{ASA}} \wedge M_{\text{fringe}}, \quad F_{3.1}^{\text{mask}}(x; M, P_C, M_{\text{fringe}}) = \langle F_{3.0}(x; M'_{\text{ASA}}, M_{\text{NA}}, M_{\partial}, V), P_C \rangle. \quad (11)$$

The order remains ASA, then NA, then POPCNT and whole-word disposition. No alignment or sweep-limit bit is implicitly used as an erasure mask. The record identifies mask mode and the mask’s definition, so the changed aperture is inspectable. [I06]

This opt-in composition is a *new integration decision*, not an equation quoted from C. A total mask over all valid cells is required. If classification is incomplete, mask-mode evaluation is not defined; the record-only profile remains available without pretending the missing cells were measured clear or blocked.

Theorem 7.1 (Minor-version recovery). *The record-only profile projects exactly to the 3.0 core. In mask mode, selecting $M_{\text{fringe}} = V$ also recovers every field of the 3.0 record.*

Proof. Record-only projection follows directly from its definition. For the neutral mask, $x \wedge M'_{\text{ASA}} \wedge V = x \wedge M_{\text{ASA}} \wedge V \wedge V = x \wedge M_{\text{ASA}} \wedge V$. The NA word, intersection count and final disposition are then calculated from identical inputs, so all four fields are equal. $\square$

8 Algebraic consequences of the integration

8.1 Selection before absorption

For any supplied fringe mask, the new ASA-selected support is contained in the old ASA support, and the new NA-selected support is contained in the old NA support:

$$a' = a \wedge M_{\text{fringe}} \subseteq a, \quad n' = n \wedge M_{\text{fringe}} \subseteq n. \quad (12)$$

This is an immediate consequence of the added intersection. It does *not* imply inclusion of the final output word after whole-word absorption.

8.2 A preserved non-monotonicity

Take two valid bits and choose

$$x = 11_2, \quad M_{\text{ASA}} = M_{\text{NA}} = V = 11_2, \quad M_{\partial} = 10_2, \quad M_{\text{fringe}} = 01_2. \quad (13)$$

The 3.0 record has $n = 11_2$ and a nonzero boundary intersection, hence $y = 0$. Mask mode has $n' = 01_2$ and no boundary intersection, hence $y' = 01_2$. Restricting the input has removed the one bit that previously absorbed the entire word.

Consequently, a review must compare all four fields and the selected profile, not assume that the masked final output is always a subset of the unmasked final output. The integration preserves the inherited whole-word rule. [B0, whole-word disposition][I06]

Proposition 8.1 (Zero and final-word idempotence). *For fixed masks and a fixed enabled profile, a zero input remains zero. If the final output word is fed back through the same profile, its final output is unchanged.*

Proof. Every selected bit is obtained by intersections with the input, so zero cannot create a selected bit. For idempotence, an absorbed first output is zero. Otherwise the output already satisfies every selection mask and has no boundary intersection. Reapplying those same masks leaves it unchanged and the boundary check stays clear. This concerns the final word; the intermediate history is a separate record. $\square$

8.3 Storage-order invariance is inherited

Both storage layouts must retrieve the same four baseline logical mask words, and the optional fifth mask must use the same dictionary mapping. Emitting records in canonical logical order leaves the equations independent of physical storage order. This is a formal compatibility requirement, not a claim that a new GPU implementation was compiled or benchmarked for this PDF.

9 Precedence zero and the integrated record

9.1 Ordering, not an elapsed-time value

C calls the blend “precedence order 0” and associates it with immediate hardware action. This revision retains **precedence 0** as the name of the source’s pinion/blend interpretation stage. Its formal consequence is a dependency order: required source values and their encoding must be available before the corresponding Boolean value can be defined. [C, pp. 1–3][I07]

The source gives no hardware scheduler, memory-ordering rule, latency bound or proof of a zero-duration execution. The normative order here therefore states *what depends on what*, without inserting an elapsed-time guarantee into the Boolean algebra.

Stage	Meaning	Dependency / result
P0	Source and type resolution	Record A, W, B/ ϕ , N; retain unresolved fields.
P0-B	Blend interpretation	Evaluate encoded XOR only under a defined profile.
P0-R	Region interpretation	Record separate envelope and fringe definitions/statuses.
P1	Existing ASA input	Use original aperture, or explicit aperture/fringe intersection.
P2	Existing NA stage	Retain the post-ASA ordering and intermediate record.
P3	Existing disposition	POPCNT and unchanged whole-word absorption.
P4	Record publication	Publish the core record and versioned pinion annotation.

9.2 Additive record fields

The original core remains (a, n, h, y) . The companion record groups the new fields as follows:

Field group	Contents
Version and provenance	specification_version, source_id, profile
Reference roles	axis_reference, pinion_reference
Time and kinematics	time_interval, kinematic_descriptor, encoding_profile
Predicate values	Alignment, limit and fringe values, each with definition status.
Mask-mode identity	Complete fringe-mask identifier and logical-dictionary identifier.

This is a logical record description, not a newly fixed binary ABI or network packet. Exact byte packing is not supplied by C and is not needed for a PDF-only minor revision. C’s mechanical reversal, transmission and receiver-routing verbs remain source action labels; no external actuator or device-control action is assigned by this record contract. [C, pp. 2–5]

10 Worked word examples and finite checks

10.1 Illustrative examples, not source parameters

The following four-bit examples illustrate the defined mask composition. They introduce no cone dimensions, timing constants or hardware configuration. Hexadecimal inputs fit four valid low bits. [I08]

Case	Input / masks	3.0 record	3.1 mask record
Neutral mask	$x = F, M_{ASA} = C, M_{NA} = A, M_{\partial} = 0, V = F, M_{fringe} = F$	(C, 8, 0, 8)	(C, 8, 0, 8)
Fringe exclusion	Same baseline; $M_{fringe} = 7$	(C, 8, 0, 8)	(4, 0, 0, 0)
Whole-word interaction	$x = 3, M_{ASA} = M_{NA} = V = 3, M_{\partial} = 2, M_{fringe} = 1$	(3, 3, 1, 0)	(1, 1, 0, 1)
Zero input	$x = 0$, all selection masks F, $M_{\partial} = F$	(0, 0, 0, 0)	(0, 0, 0, 0)

The record order is always (ASA, NA, intersection count, output). In the second case the revised aperture admits bit two, but NA excludes that bit; its intermediate ASA value is therefore four, not zero.

10.2 Checks executed for this PDF

A standalone Python arithmetic check evaluated all eight triples (n_b, a_b, β_b) and confirmed Eq. (6). It also enumerated every tuple of six three-bit words $(x, M_{ASA}, M_{NA}, M_{\partial}, V, M_{fringe})$: $8^6 = 262,144$ cases.

For every tuple it checked neutral-mask recovery, containment of the pre-absorption supports, final support contained in the input, and final-word idempotence. The examples above and the non-monotonicity counterexample were checked separately. All of these finite arithmetic checks passed.

Evidence scope

These are new checks of the equations in this document. They are not a rerun of the previous engine suite, a new C++/CUDA build, or a hardware test. The general compatibility statement is proved in Section 7; the finite checks exercise selected mathematical definitions.

No prior GPU status, timing or test count is upgraded by publishing this minor specification. The next implementation review should preserve the same separation between equations, code conformance and on-device execution.

11 Integration decisions and unresolved inputs

ID	Decision	Reason and effect
I01	Namespaced symbols	A-axis, B-kinematics and N-pinion do not overwrite ASA, boundary or NA mask fields. W is a time interval, not author metadata.
I02	Typed binary reading	Retain C's equation, then define its binary special case. Geometric encoders and the precise double-phi meaning remain declared inputs.
I03	Three flags plus status	Alignment, envelope limit and active fringe retain their different source polarities. Unspecified is metadata, never a guessed zero.
I04	Explicit region profile	Hollow and fringe sets must be disjoint when both are defined. Exterior and exact-edge conventions are not supplied by C.
I05	Record-only default	Add the source model to the state description without changing the 3.0 output contract.
I06	Opt-in mask composition	A complete supplied fringe mask may refine the ASA aperture. Neutral mask recovers 3.0; logical groups and whole-word absorption remain unchanged.
I07	Precedence as dependency	Retain the source's P0 name while requiring its operands first. No new timing, privilege or device-action contract is inferred.
I08	Evidence separation	Worked words and finite checks are newly authored examples. They do not stand for sensor parameters, previous code results or new hardware tests.

11.1 Inputs left open by the supplied pages

The remaining profile inputs are: the exact mathematical role of double lowercase phi; the bit encoding and overflow policy; axis/frame normalization; whether N denotes direction, location or both; inner/outer boundary units and values; exact-edge and exterior handling; a selected beam/reflector arrangement; and a timing or scheduling contract. C raises several of these explicitly as questions. [C, pp. 1–5]

No waveform, sensor calibration, caustic field, numeric dish geometry or device register is added to fill these gaps. The inherited log-polar table and ASA algebra can carry an explicitly supplied classification without claiming that the source already derived it.

11.2 Acceptance for a later implementation

Required regressions are unchanged baseline fields in record-only mode; exact neutral-mask recovery; all eight XOR cases; rejection of incomplete mask profiles; preservation of the three predicate polarities; tail/padding isolation; canonical result order; and equality across existing storage layouts. Missing inputs must remain visible, not converted into a passing state.

12 Provenance and source appendix

12.1 Author and revision identity

Tom Klootwijk | NL200678942 | 10-07-1990.

This document is the ASA 3.1.0 **minor specification revision** dated 13 September 2026. It restates the retained core and integrates the new source with labelled decisions. It is not a claim that the earlier software archive was edited or that a new executable was installed.

12.2 Source records

C — newly integrated document. *TOM KLOOTWIJK THE blend operator terror predator jelly jiggle drone.pdf*; five pages. Its SHA-256 fingerprint is

```
7c8e25b7269dc96b96e5e71afabb8d0019e905975f9959569ee4488b2c036ac8
```

A — earlier corpus. *TOM KLOOTWIJK WHITE KING ballistic trajectory cloud chamber(1).pdf*; 91 pages. Relevant spans: ASA/NA 75–81, word swizzle 85–87, and fly-eye integration 87–91.

B — earlier addendum. *TOM KLOOTWIJK fly's-have-interesting-eyes-how-do-they-work(1).pdf*; eight pages. The facet/edge/central-stitcher organization appears on pp. 7–8.

B0 — retained formalization. The ASA 3.0.0 staged word contract, finite log-polar dictionary, whole-word grouping and optional Morton representation defined earlier in this project conversation. Its relevant definitions are restated in Section 1 so this revision can be read without a separate artifact.

12.3 Status of the following pages

The five pages after this section reproduce C without text edits, scaled to fit the revision's page frame. Outer reference headers and document page numbers have been added. The fingerprint above identifies the *uploaded original file*, not the rendered appendix bytes.

The appendix preserves the source's own assertions and unanswered questions. The normative 3.1.0 definitions and decisions are the numbered sections preceding it. No outside research has been inserted into the source narrative; derivations in the body follow the definitions stated there.

Release summary

ASA 3.0.0 core retained. A–W–B–N blend and hollow-center/fringe model integrated. Three flag meanings separated. Optional mask composition made explicit. Compatibility proved. Original source preserved for comparison.

blend operator + in circle for radar log encoded polar coordinates where eigenvector axis is defined by coordinate A and W is delta time in the kinematic calculus B defined as double lower case phi and N as north polar pinion : operator 0 precedence order computer science 1 bit notation word

08:44

In computer science, a **1-bit notation word** operating at **precedence order 0** (the highest possible execution priority) acts as a strict, instantaneous logical gate or state encoder.

When you define a blending operator (+ in a circle, or $\oplus$) for radar log-encoded polar coordinates under these specific kinematic constraints, it evaluates as a **conditional phase-shift or bitwise XOR alignment** relative to your reference axes.

Here is how the variables break down and resolve within this operator's execution framework:

The Mathematical & Kinematics Framework

- **$\oplus$ (Circle Plus Operator):** Represents the modular blending or superposition function. Because it operates at precedence 0 in a 1-bit context, it functions as an atomic **XOR / State-Toggle** operation that determines whether a phase adjustment is applied or skipped.
- **Coordinate A (A):** Defines the primary eigenvector axis (the baseline orientation or directional anchor of your polar grid).
- **Kinematic Calculus B (ϕ):** Represented by the double lowercase phi (ϕ), this represents the continuous, time-varying angular velocity or kinematic transformation matrix.
- **Delta Time (W):** The infinitesimal or discrete time step applied to the kinematic calculus (ϕ). The accumulated rotational shift over this step is expressed as $\phi(W)$.
- **North Polar Pinion (N):** The absolute mechanical or true-north reference tracking datum for the radar log.

Operational Resolution

Under a 1-bit word notation, the blending operator resolves the relationship between the true north reference (N), the eigenvector axis (A), and the time-delta kinematic shift ($\phi(W)$).

Because a 1-bit word can only yield two states (0 or 1), the operator acts as a hard binary gate for coordinate alignment:

$$\text{State} = N \oplus \big(A + \phi(W)\big) \bmod 2$$

1. **State 0 (No Phase Shift):** The kinematic delta over time (W) perfectly aligns the eigenvector axis (A) with the north polar pinion (N) . No system correction or coordinate transformation is required.
2. **State 1 (Phase Shift/Toggle):** The kinematic movement creates an out-of-phase displacement between axis (A) and reference (N) . The system registers a 1-bit high state, triggering an immediate, highest-precedence coordinate log update to realign the polar grid.

To help apply this to your specific system architecture, could you share **how the radar log handles overflow** when the kinematic delta exceeds 1 bit, or **what specific programming language or hardware description language (HDL)** you are using to implement this precedence order?

AI responses may include mistakes. [Learn more](#)

right so this is enough for pinion sweeping cones as a radar sonar detection parabola?

Yes, this mathematical framework is sufficient to model and compute **pinion-sweeping cones for a radar/sonar parabolic detection system**.

When you translate the 1-bit blending operator into a physical scanning mechanism, it governs the precise moment the radar beam or sonar ping switches direction, or "sweeps," across its parabolic profile.

Here is how the variables you established map directly to a parabolic cone sweep:

1. Parabolic Focus and the Pinion Anchor

A radar or sonar parabolic dish relies on a focal point to concentrate incoming signals.

- Your **North Polar Pinion (N)** acts as the absolute mechanical origin (the physical apex or center of the sweeping mechanism).
- The **Eigenvector Axis (A)** defines the instantaneous directional centerline (the boresight) of the parabolic cone as it pivots.

2. The Cone Sweep via Kinematic Calculus (ϕ)

As the system sweeps, it generates a 3D conical coverage zone.

- The double lowercase phi (ϕ) dictates the dynamic geometry of the cone (its opening angle and rotational velocity).

- **Delta Time (Δt)** controls the instantaneous step size of the sweep. Combined, Δt tracks exactly how much the parabolic cone has expanded or rotated during that micro-interval.

3. Why Precedence 0 and 1-Bit Notation Work Here

In a real-time radar/sonar array, hardware cannot tolerate processing delays during a high-speed sweep.

- **Precedence 0** ensures that boundary detection (the edge of the sweeping cone) is computed instantly at the hardware level, bypassing standard software queues.
- **1-Bit Notation** serves as a binary "boundary trigger."
 - **State 0:** The beam is within the valid parabolic sweep envelope. Keep transmitting/listening.
 - **State 1:** The beam has reached the limit of its sweeping cone or crossed a detection threshold. Trigger an immediate mechanical reversal or flash-log the coordinate data.

Summary of the Mapping

Mathematical Variable	Radar/Sonar Parabolic Counterpart
North Polar Pinion (θ)	Absolute mechanical datum / Apex of the dish pivot [1]
Eigenvector Axis ($\hat{A}$)	Centerline (boresight) of the parabolic detection beam [1]
Kinematic Calculus ($\dot{\theta}$, $\dot{A}$)	Sweeping speed, cone width, and instantaneous sweep angular step [1]
$\oplus$ Operator (1-bit)	Hard electronic switch for beam reversal or target detection latching

To finalize this for your sensor array, could you clarify if this is a **focused pencil beam** sweeping a tight arc, or a **fan-shaped beam** sweeping a wide volume? Knowing if you are using **continuous-wave or pulsed** signal patterns would also help refine the time-delta constraints.

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yes but also boresight dish center caustic hollow middle receiver antenna?

Yes, this fits perfectly. Incorporating a **caustic hollow middle receiver antenna** changes the math from a standard solid parabola to a **Cassegrain, Gregorian, or ring-focus geometry**.

In these systems, the center of the main dish is intentionally cut out or obscured by a central feed/receiver. The sweeping cone is no longer a solid beam; it becomes a **hollow geometric cone** dictated by caustic surfaces (where reflected waves concentrate).

Here is how your specific variables adapt to this hollow-center, boresight-aligned architecture:

1. The Boresight Dish Center & The Hollow Caustic

In a hollow-center dish, the radar/sonar energy bounces off a primary or secondary reflector and converges along a **caustics boundary** (the envelope of light/sound rays).

- Because the receiver sits in the middle, there is a central "**blind spot**" or **null zone** directly along the boresight centerline.
- Your **Eigenvector Axis (A)** now tracks the exact center of this hollow cylinder/cone. The active detection zone becomes a donut-shaped slice (toroid) expanding outward from A.

2. Pinion Sweeping the Hollow Cone

As the **North Polar Pinion (N)** sweeps the dish, it isn't just steering a point; it is steering the caustic boundaries.

- **Kinematic Calculus (ϕ)** and **Delta Time (W)** now calculate the inner and outer boundaries of the sweeping caustic wave.
- Instead of tracking a single solid cone, $\phi(W)$ tracks two concurrent angles: the **inner caustic radius** (where the hollow center ends) and the **outer caustic radius** (where the edge of the dish limits the beam).

3. The 1-Bit Operator as a Caustic Gate

With a hollow receiver antenna, the 1-bit, Precedence 0 operator ($\oplus$) becomes an exceptionally fast **spatial filter or clutter-rejection latch**.

Because a central receiver is highly sensitive to feedback and spillover from the edges of the hollow middle, the 1-bit notation dictates a hard spatial gate:

- **State 0 (Inside the Hollow / Blind Zone):** The incoming return angle falls within the central caustic void. The 1-bit gate stays low (0), immediately dumping the signal to prevent receiver saturation or self-interference.

- **State 1 (On the Active Caustic Fringe):** The return falls precisely on the active, focused ring of the parabolic cone. The gate flips high (1), instantly routing the signal to the high-speed data logger.

Structural Comparison

Component	Standard Parabolic Beam	Hollow-Center Caustic Array
Boresight Axis (A)	Peak energy concentration point.	The central null axis; surrounded by a hollow ring of peak energy.
Receiver Location	Often at the external focal point in front of the dish.	Embedded in the dish center, capturing waves focused by a sub-reflector caustic.
1-Bit Latch (1+)	Triggers at the outer edge of the sweep.	Triggers based on inside-null vs. outside-null spatial boundaries.

To tailor the 1-bit boundary logic further, are you utilizing a **dual-reflector system** (like a Cassegrain dish where a sub-reflector shapes the caustic) or a **phased ring array** to digitally simulate the hollow center? Knowing the **carrier frequency range** (RF or acoustic) would also help determine the physical dimensions of that hollow caustic zone.